

Aman Modanwal

Assistant Software Engineer

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Greater Noida, India

Profile

Software Engineer with **3.5+ years of experience** in designing and developing full-stack web applications and desktop solutions. Proficient in **React.js, Node.js, Express.js, JavaScript, HTML, CSS, MySQL, and Electron**. Experienced in building REST APIs, real-time applications, and responsive user interfaces. Strong knowledge of **Linux, Git, Networking (TCP/IP, DNS), ONVIF, and Network Video Recorder (NVR) systems**. Passionate about solving complex problems, delivering high-quality software, and collaborating in agile development environments.

Professional Experience

Asst. Software Engineer

PPS INTERNATIONAL Pvt Ltd

01/2023 – present
Greater Noida, India

- Developed and maintained full-stack web and desktop applications using **React.js, Node.js, Express.js, Electron, and MySQL**.
- Designed and integrated **RESTful APIs**, implemented **JWT-based authentication**, and optimized backend performance.
- Built responsive and reusable UI components using **React.js**, improving user experience and application maintainability.
- Developed real-time features using **WebSocket/Socket.IO** for instant data synchronization and communication.
- Worked on **ONVIF-based IP camera integration**, real-time monitoring, and **Network Video Recorder (NVR)** applications.
- Performed debugging, testing, code reviews, and production issue resolution in **Linux** environments.
- Collaborated with cross-functional teams using **Agile/Scrum** methodologies to deliver features within sprint timelines.
- Prepared technical documentation, API documentation, and supported software maintenance and enhancements.

Diploma Engineering Trainee (DET)

PT Communications System Pvt Ltd

12/2022 – 01/2023
Greater Noida, India

- Contributed to software development and testing activities, including feature implementation, unit testing, and debugging.
- Worked with relational and/or NoSQL databases for data modeling, query optimization, and efficient data retrieval.
- Gained hands-on experience in system configuration, API integration, and software module integration within existing application architecture.

Technical Skills

- Frontend:** React.js, JavaScript (ES6+), Angular, TypeScript, HTML5, CSS3, SCSS, Redux Toolkit, React Hooks, Context API, Bootstrap, React Bootstrap, Chakra UI, jQuery
- Backend:** Node.js, Express.js, REST APIs, Django, JWT Authentication, WebSocket, Socket.IO
- Languages:** JavaScript, TypeScript, Python, basic C , Kotlin
- Databases:** SQL Server MySQL, MongoDB
- Tools & Technologies:** Electron, Electron Builder, Vite, Webpack, Babel, GitHub, NPM, Yarn
- Networking & Protocols:** TCP/IP, DHCP, DNS, RTSP, ONVIF, SOAP/XML
- Operating Systems:** Linux, Windows
- Cloud & DevOps:** AWS (EC2, S3, RDS), Git, Docker

Projects

Network Video Recorder (NVR)

Real-Time Video Monitoring System

- Developed a web-based NVR application for monitoring and managing ONVIF IP cameras.
- Built the frontend using **React.js** and the backend using **Node.js** and **Express.js**.
- Implemented camera discovery, live video streaming, and recording management.
- Developed REST APIs for camera management and user authentication.
- Integrated **ONVIF (SOAP/XML)** and **RTSP** for camera communication and live streaming.

SPUT Tool

Automated IP Monitoring and Content Deployment Tool

- Designed a tool to **list and monitor IoT Web Server Board** modules by continuously pinging IPs, displaying real-time health indicators (tick or cross) to ensure active status.
- Integrated folder validation to verify authorized content selection, preventing errors with clear feedback for incorrect choices.
- Streamlined **data transfer to selected IPs or multiple Devices** through a user-friendly interface, ensuring efficient and accurate content distribution.

IOB Server

Integrated Content and Service Management Platform for Railway Passengers

- Developed a **user-friendly frontend** with dedicated sections for movies, music, videos, kids' zone, journey details, tourist places, and service requests.
- Implemented secure user authentication to ensure data privacy and control user access.
- Enabled seamless service request submission and management for journey-related services, enhancing passenger convenience.

Education

BTech, CSE

Dr A P J Abdul Kalam Technical University

2024 – present

Meerut, India

Full Stack Development

Apna College

01/2023 – 08/2023

Hybrid, India

Diploma, CSE(74%)

Board of Technical Education

2019 – 2022

Jaunpur, India

Certificates

Problem Solving (Basic)  — HackerRank

DSA with JAVA — APNA COLLEGE

Master Javascript — GeeksforGeeks

Languages

• Hindi

• English